

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 4: Approach & En-route Communications

Syllabus Source: CAE Oxford (pp. 89-118) | Pass Target: >= 90% AviationExam

1. IFR DEPARTURE CLEARANCES & LEVEL INSTRUCTIONS

Initial IFR clearances contain route of flight, assigned SID, initial altitude/FL, and transponder squawk code:

- Clearance Delivery: 'IBERIA 123, CLEARED TO PARIS CHARLES DE GAULLE VIA PINUS 1A DEPARTURE, CLIMB FLIGHT LEVEL 80, SQUAWK 4215'.
- Climb & Descent Phrasing: 'CLIMB TO FLIGHT LEVEL 240', 'DESCEND TO 4000 FEET QNH 1018'.
- Transition Altitude & Level: Transition Altitude (TA) is referenced to QNH (feet). Transition Level (TL) is referenced to standard pressure 1013.25 hPa (Flight Levels). Transition Layer is the airspace between TA and TL.

2. STANDARD EN-ROUTE POSITION REPORTING

Unless operating under radar coverage where 'Radar Contact' has been confirmed and position reports waived, an IFR flight must transmit standard position reports over designated compulsory reporting points:

Report Element	Description	Standard Radiotelephony Phrasing
1. Aircraft Identification	Callsign	'IBERIA 123'
2. Position	Compulsory reporting fix	'OVER CLACTON'
3. Time	Minutes past the hour	'AT 25'
4. Flight Level / Altitude	Present level	'FLIGHT LEVEL 280'
5. Next Position & Estimate	Next fix and estimated time	'ESTIMATING OTFORD AT 42'
6. Ensuing Significant Point	Subsequent fix name	'NEXT LYDD'

3. RADAR VECTORING & SPEED CONTROL COMMANDS

- Vectoring Instructions: 'TURN LEFT HEADING 090', 'LEAVE HEADING 270 FLY HEADING 310', 'FLY HEADING 040 FOR INTERCEPT'.
- Speed Control: 'MAINTAIN 250 KNOTS', 'REDUCE TO MINIMUM CLEAN SPEED', 'RESUME NORMAL SPEED'.
- Intercepting Localizer: 'CLEARED ILS APPROACH RUNWAY 24, REPORT ESTABLISHED ON LOCALIZER'.

4. SURVEILLANCE RADAR APPROACH (SRA)

A Surveillance Radar Approach (SRA) is an instrument approach guided by a radar controller transmitting heading instructions and advisory altitudes per nautical mile from touchdown:

[!] EXAM TRAP (AVIATIONEXAM) - SRA Transmission Intervals & Missed Approach Criteria (Exam Trap)

- Transmission Frequency on Final Approach: The radar controller must transmit to the aircraft at intervals of NOT MORE THAN 5 SECONDS during final approach! - Missed Approach Rule: If no transmission is received from the controller for an interval of 5 SECONDS (or 15 seconds in some national procedures), the pilot MUST IMMEDIATELY EXECUTE A MISSED APPROACH! - Termination of SRA: The controller terminates the approach at 2 NM from touchdown (or 1/2 NM for high-precision SRA), or when the pilot reports the runway in sight.

5. CONTROLLER-PILOT DATA LINK COMMUNICATIONS (CPDLC)

CPDLC allows direct two-way digital text messaging between ground ATC automation and the aircraft FMS. Benefits include eliminating voice frequency congestion, accents, and garbled transmissions. Standard crew responses to CPDLC uplink clearances are: WILCO, UNABLE, STANDBY, or ROGER.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What are the required elements of a standard ICAO position report in that exact sequence?**

[A] Callsign, Altitude, Speed, Fuel, Next point [B] Callsign, Position, Time, Flight Level, Next position and estimated time, Ensuing significant point [C] Callsign, Heading, Wind, Estimated time of arrival [D] Callsign, Squawk code, Distance to go

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The mandatory sequence is: Callsign -> Position -> Time -> Flight Level -> Next position & estimate -> Ensuing point.

Q2: During a Surveillance Radar Approach (SRA), what action must the pilot take if radio contact with the radar controller is lost?

[A] Continue descent to MDA and land visual [B] Immediately execute a missed approach [C] Switch to tower frequency and land [D] Orbit in place

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. On SRA final approach, communications failure (> 5s silence) mandates an immediate missed approach climb.

Q3: At what approximate distance from touchdown is a standard Surveillance Radar Approach (SRA) terminated?

[A] 10 NM [B] 5 NM [C] 2 NM (or 1/2 NM if high-precision) [D] At the runway threshold

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Standard SRA terminates at 2 NM from touchdown (or 0.5 NM if certified to lower minima).

Q4: What phrase does a pilot transmit upon becoming established on the ILS localizer course?

[A] 'Localizer captured' [B] 'Established on localizer' (or 'Established runway 27') [C] 'On the beam' [D] 'ILS locked'

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. 'ESTABLISHED' or 'ESTABLISHED ON LOCALIZER' is the standard ICAO radiotelephony report.

Q5: What does the instruction 'RESUME NORMAL SPEED' signify to a flight crew following ATC speed restriction?

[A] Accelerate to maximum speed V_{mo} [B] The previously issued speed restriction is cancelled; pilot flies normal operational flight plan speeds [C] Fly exactly 250 kt [D] Slow down to approach speed

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. 'Resume normal speed' cancels ATC-imposed speed control, returning aircraft to operator's flight plan profile.

Q6: What is the primary operational objective of CPDLC in oceanic and en-route airspace?

[A] Replacing visual scans [B] Eliminating voice frequency congestion and audio language misunderstandings through digital text messaging [C] Automatically controlling aircraft engines [D] Providing weather radar pictures

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. CPDLC replaces voice communications with structured data messages, reducing HF/VHF frequency congestion.